

Computability & Logic – Exam June 2019 (corrected)

ID

Name

Each of the following 20 questions has 4 possible answers of which exactly one is correct. For each question, you may select one or more answers by checking the appropriate boxes. Your test is graded as follows:

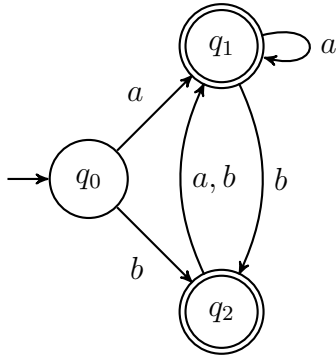
- If you select only the correct answer, you receive 2 points.
- If you select 2 answers, one of which is correct, you receive 1 point.
- If you select 3 answers, one of which is correct, you receive 0.41 points.
- If you select no answer or all answers, you receive 0 points.
- If you select only one answer, and it is wrong, you receive -0.67 points.
- If you select 2 answers, that are both wrong, you receive -1 point.
- If you select 3 answers, that are all wrong, you receive -1.25 points.

Note that perfect answers yield a score of 40 and random guessing may give you a negative score. For the logical basis of this score system, see Frandsen and Schwartzbach, *A singular choice for multiple choice*.

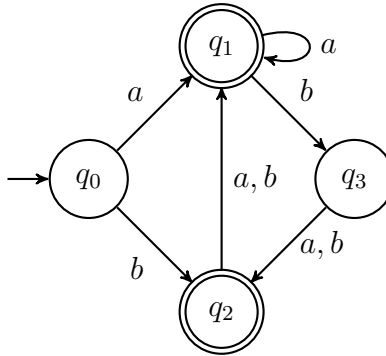
Note: this is an open-book exam. You may use books, lecture slides and other notes. The use of electronic devices is *not* permitted.

Question 1

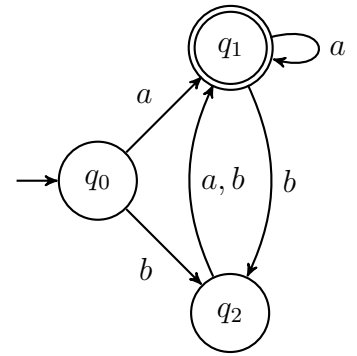
Which of these (deterministic) finite automata are *minimal*?



(M_1)



(M_2)

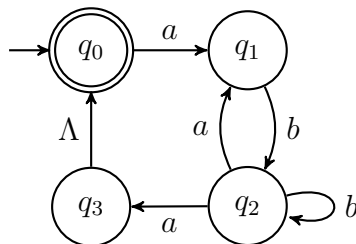


(M_3)

- a ☐ All of them are minimal
 b ☐ Only M_1 and M_2 are minimal
 c ☒ Only M_2 and M_3 are minimal
 d ☐ Only M_3 is minimal

Question 2

Consider the following NFM M :



Which of the following regular expressions does *not* describe the same language as M ?

- a ☐ $\left(a(bb^*a)^*bb^*a \right)^*$
 b ☐ $\left(ab(b^*ab)^*b^*a \right)^*$
 c ☒ $\left(abb^*(ab)^*a \right)^*$
 d ☐ $\left(ab(ab + b)^*a \right)^*$

Question 3

Lars Mads Jensen wants to use the pumping lemma, to prove that some language L is not regular. Which of the following proof sketches is the most promising start?

- a ☐ Let k be arbitrary, choose $z := a^k b^k c^k \in L$ Let $u := a^k, v := b^k, w := c^k$. Clearly, $z = uvw$ and ... Then for $i := 3$, $uv^i w \notin L$.
- b ☒ Let k be arbitrary, choose $z := a^k b^k c^k \in L$ Let u, v, w be arbitrary words, such that $z = uvw$ and ... Then for $i := 3$, $uv^i w \notin L$.
- c ☐ Let $k := 10$, let z be an arbitrary word in L of length k . Let u, v, w be arbitrary words, such that $z = uvw$ and ... Then for $i := 3$, $uv^i w \notin L$.
- d ☐ Let k be arbitrary, choose $z := a^k b^k c^k \in L$ Let u, v, w be arbitrary words, such that $z = uvw$ and ... Then, for arbitrary $i \geq 0$, $uv^i w \notin L$.
-

Question 4

Which of these grammars is in Chomsky Normal Form?

a ☒
$$\begin{array}{l} \hline S \rightarrow AB \\ A \rightarrow AS \mid a \\ B \rightarrow a \mid b \mid AA \\ \hline \end{array}$$

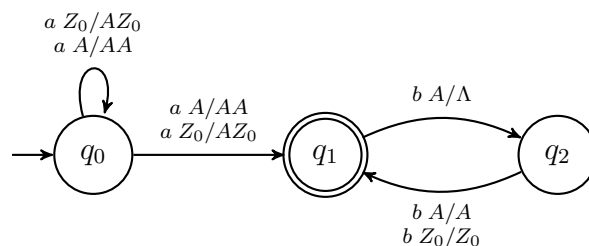
b ☐
$$\begin{array}{l} \hline S \rightarrow AB \\ A \rightarrow AS \mid a \\ B \rightarrow A \mid b \\ \hline \end{array}$$

c ☐
$$\begin{array}{l} \hline S \rightarrow AB \\ A \rightarrow AS \mid \Lambda \\ B \rightarrow a \mid b \\ \hline \end{array}$$

d ☐
$$\begin{array}{l} \hline S \rightarrow AB \\ A \rightarrow AS \mid a \\ B \rightarrow a \mid b \mid ASA \\ \hline \end{array}$$

Question 5

Consider the following Push-Down Automaton M , with initial stack symbol Z_0 . Consider $\mathcal{L}(M)$, the language accepted by M (by final state).



Which of these grammars generates exactly the same language $\mathcal{L}(M)$?

a ☐
$$\begin{array}{l} S \rightarrow aT \\ T \rightarrow aTbb \mid \Lambda \end{array}$$

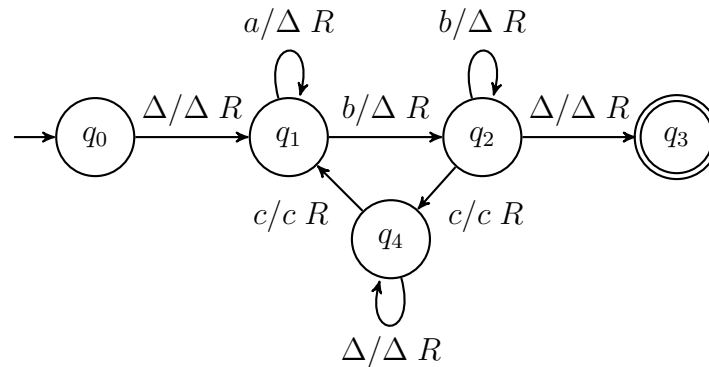
b ☐
$$\begin{array}{l} S \rightarrow aTbb \\ T \rightarrow aTbb \mid \Lambda \end{array}$$

c ☒
$$\begin{array}{l} S \rightarrow aTbb \mid aS \mid a \\ T \rightarrow aTbb \mid \Lambda \end{array}$$

d ☐
$$S \rightarrow aS \mid Sbb \mid a$$

Question 6

Consider the following Turing Machine:



- ☒ This TM accepts the language $a^*bb^*(cca^*bb^*)^*$
☐ This TM decides the language $a^*bb^*(cca^*bb^*)^*$
☐ This TM accepts the language $(cc)^*$
☐ This TM decides the language $(cc)^*$

Question 7

Consider the following statement:

(*) L is a recursively enumerable language

Which of the following statements is equivalent to the previous statement?

- (1) there is a TM that accepts precisely all $w \in L$
(2) there is a TM that accepts all $w \in L$ and rejects all $w \notin L$
(3) there is a TM that accepts all $w \in L$ and hangs on all $w \notin L$

- ☐ (*) is equivalent to (1) and (2), but not to (3)
☒ (*) is equivalent to (1) and (3), but not to (2)
☐ (*) is equivalent to (2) and (3), but not to (1)
☐ (*) is equivalent to (1) only, but not to (2) or (3)

Question 8

Finish the sentence in the only correct way:

Every Context-Free Language can be recognized by ...

- ☒ ...some Push-Down Automaton
☐ ...some Deterministic Push-Down Automaton
☐ ...some Non-deterministic Finite Automaton
☐ ...the product of a finite number of NFAs

Question 9

Only one of the following statements holds. Which one?

The set of valid statements of ...

- a ☐ ... arithmetic is recursive
- b ☐ ... arithmetic is recursively enumerable
- c ☐ ... first-order predicate logic is recursive
- d ☒ ... first-order predicate logic is recursively enumerable

Question 10

Consider the following formula of propositional logic: " $p \rightarrow (\neg p \wedge q)$ ". Which of the following statements about this formula is true?

- a ☒ the formula is satisfiable, but not valid
- b ☐ the formula is satisfiable and valid
- c ☐ the formula is valid, but not satisfiable
- d ☐ the formula is neither valid, nor satisfiable

Question 11

Consider the following formula of predicate logic: " $(\forall x(P(x) \rightarrow Q(x))) \wedge a = b \wedge P(a) \wedge \neg Q(b)$ ". Which of the following statements about this formula is true?

- a ☐ the formula is satisfiable, but not valid
- b ☐ the formula is satisfiable and valid
- c ☐ the formula is valid, but not satisfiable
- d ☒ the formula is neither valid, nor satisfiable

Question 12

Which of the following statements on SAT (the satisfiability problem in propositional logic) is not known to be true?

- a ☐ SAT is in NP
- b ☐ SAT is NP-hard
- c ☐ SAT is NP-complete
- d ☒ SAT is in P

Question 13

Which of the following problems is *not* decidable?

- a ☐ The TM M halts within at most 10 steps on input Λ
- b ☐ The TM M never writes into an empty cell on some given input w
- c ☒ The language of TM M contains a word of length 10
- d ☐ The language of TM M is recursively enumerable

Question 14

Which statement is wrong?

- ☐ *a* The class of regular languages is closed under intersection
- ☐ *b* The class of recursive languages is closed under intersection
- ☒ *c* The class of context-free languages is closed under intersection
- ☐ *d* The class of polynomial languages (P) is closed under intersection

Question 15

Consider the list of proof rules of natural deduction for predicate logic:

$$(premise), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2, \rightarrow i, \rightarrow e, \neg e, \neg i, \neg\neg e, \forall e, \forall i, \exists e, \exists i, \perp e.$$

The sequent " $r, \forall x((p \rightarrow r) \rightarrow q(x)) \vdash q(a)$ " can be proved using the proof rules:

- ☒ *a* $premise, \forall e, \rightarrow i, \rightarrow e$
- ☐ *b* $premise, \forall i, \rightarrow i, \rightarrow e$
- ☐ *c* $premise, \forall i, \forall e, \rightarrow e$
- ☐ *d* this sequent is not valid, so it cannot be proved

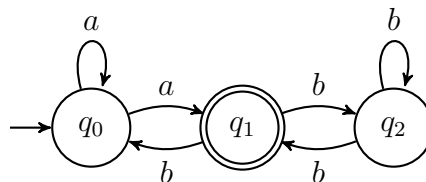
Question 16

Which of these statements about verification is wrong?

- ☐ *a* Model checking is a fully automatic procedure
- ☐ *b* Program verification typically requires annotation with invariants
- ☐ *c* SAT solving is a fully automatic procedure
- ☒ *d* SAT solving requires solving arbitrary arithmetic formulas

Question 17

Consider the following NFA:



We apply the subset-construction to this NFA, in order to obtain a deterministic FA, *without any minimisation steps*. How many reachable accepting states does the resulting automaton have?

- ☐ *a* 1 accepting state
- ☐ *b* 2 accepting states
- ☒ *c* 3 accepting states
- ☐ *d* 4 accepting states

Question 18

Given are three languages L_1, L_2, L_3 , with the properties:

- L_1 is a regular language
- L_2 and L_3 are context-free languages

Which of the following languages is *not necessarily* context-free?
(X' denotes the complement of set X)

- a ☐ $L_1 \cup L_2 \cup L_3$
 b ☒ $L_1 \cap L_2'$
 c ☐ $L_2 \cap L_1'$
 d ☐ $(L_2 \cup L_3) \cap L_1$

Question 19

Consider the following instance of Post's Correspondence Problem (PCP),
 $\{(010, 0), (00, 100), (11, 01)\}$:

010	00	11
0	100	01

- a ☐ This instance is solvable, even without reusing any pieces
 b ☒ This instance is solvable, but at least one piece must be reused
 c ☐ This instance is not solvable at all
 d ☐ The variant of PCP without reusing pieces is already undecidable

Question 20

Given are two languages $L_1 \subseteq \Sigma^*$ and $L_2 \subseteq \Sigma^*$ over alphabet Σ , and a TM M that computes a function $f : \Sigma^* \rightarrow \Sigma^*$. Function f translates elements in $L_1 \subseteq \Sigma^*$ to elements in $L_2 \subseteq \Sigma^*$, and elements not in L_1 to elements not in L_2 . On inputs of length n , M will always terminate in at most 2^n steps. What can we derive from this information?

- a ☒ $L_1 \leq L_2$
 b ☐ $L_2 \leq L_1$
 c ☐ $L_1 \leq_P L_2$
 d ☐ $L_2 \leq_P L_1$